

Fig. 3: Actual-size PCB layout of RF section

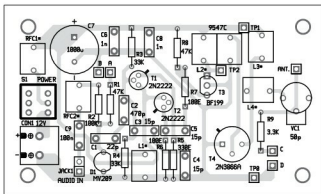


Fig. 4: Component layout of PCB of RF section

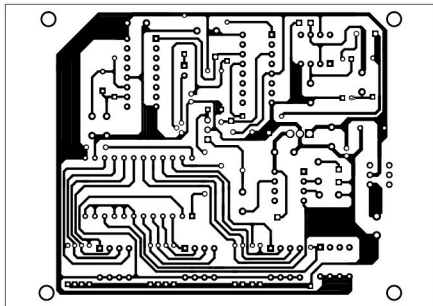


Fig. 5: Actual-size PCB layout of PLL control section

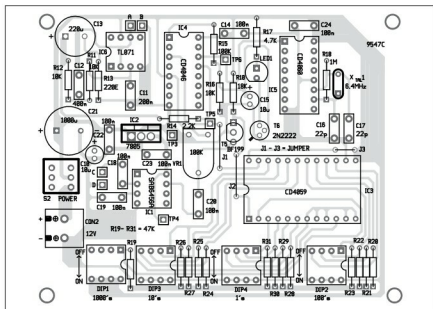


Fig. 6: Component layout of PCB of PLL control section

shown in Fig. 2), then the output will be $1.5625 \times 1024 \times 64 = 102.4\text{MHz}$.

Similarly, a DIP setting of 1000 gives us an output frequency of 100MHz. LED1 lights up to indicate a lock on the selected frequency.

Construction and testing

An actual-size, single-layer PCB layout for Unit 1 is shown in Fig. 3 and its component layout in Fig. 4. Single-layer, actual-size PCB layout for Unit 2 is shown in Fig. 5 and its component layout in Fig. 6.

Assemble the circuits on the PCBs to save time and minimise assembly errors. Unit 1 is connected to Unit 2 by a co-axial cable of short length. Open end of resistor R1, indicated as 'A' in Unit 1, is connected to pin 6 of IC6, indicated as 'A' in Unit 2, by co-axial cable. Similarly, open end of resistor R9, indicated as 'C' in Unit 1, is connected to open end of C18, indicated as 'C' in Unit 2, by co-axial cable. Ground the shield wire of the coaxial cable.

Keep all leads as short as possible. To test the circuit for proper functioning, connect stabilised 12V supply to both the units. The circuit will accept audio signal from just about anything (CD, tape, iPod or computer) and transmit the signal, which can be received by an FM radio. An external microphone amplifier can be used for speech. A good matching 50-ohm ground-plane antenna will greatly enhance the range of transmission. Use appropriate IC bases on the PCB. ●

The author is an electronics hobbyist and a small-business owner in Albany, New York, USA. His interests include designing RF circuits